



JOINT CANADA-UNITED STATES
NATIONAL STANDARD

ANSI/CAN/UL 2900- 2-3:2023

STANDARD FOR SAFETY

Software Cybersecurity for Network-
Connectable Products, Part 2-3:
Particular Requirements for Security
and Life Safety Signaling Systems



ANSI/UL 2900-2-3-2023



SCC FOREWORD

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UL Standard for Safety for Software Cybersecurity for Network-Connectable Products, Part 2-3: Particular Requirements for Security and Life Safety Signaling Systems, ANSI/CAN/UL 2900-2-3

First Edition, Dated January 31, 2020

Summary of Topics

This revision for ANSI/CAN/UL 2900-2-3 dated, September 21, 2023, includes the following changes in requirements:

– Updates to UL 2900-2-3 Based on Changes to UL 2900-1; [Table 5.1](#) (5.1), [Table 8.1](#) (8.4 and 8.5), [Table 13.1](#) (13.4 and 13.6), [Table 14.1](#), Section [14A](#), [Table 18.1](#) (Clause 18.3), [Table 19.1](#), [Table 20.1](#) (20.3 – 20.5), and [A2](#)

Text that has been changed in any manner or impacted by ULSE's electronic publishing system is marked with a vertical line in the margin.

The new and revised requirements are substantially in accordance with Proposal(s) on this subject dated June 16, 2023.

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ANSI/UL 2900-2-3-2023



JANUARY 31, 2020

(Title Page Reprinted: September 21, 2023)

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ANSI/CAN/UL 2900-2-3:2023

**Standard for Software Cybersecurity for Network-Connectable Products,
Part 2-3: Particular Requirements for Security and Life Safety Signaling
Systems**

First Edition

January 31, 2020

This ANSI/CAN/UL Standard consists of the First Edition including revisions through September 21, 2023.

The most recent designation of ANSI/UL 2900-2-3 as an American National Standard (ANSI) occurred on September 21, 2023. ANSI approval for a standard does not include the Cover Page, Transmittal Pages, Title Page, Preface or SCC Foreword.

This standard has been designated as a National Standard of Canada (NSC) on September 21, 2023.

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Preface

This is the First Edition of the ANSI/CAN/UL 2900-2-3, Standard for Software Cybersecurity for Network-Connectable Products, Part 2-3: Particular Requirements for Security and Life Safety Signaling Systems.

ULSE is accredited by the American National Standards Institute (ANSI) and the Standards Council of Canada (SCC) as a Standards Development Organization (SDO).

This Standard has been developed in compliance with the requirements of ANSI and SCC for accreditation of a Standards Development Organization.

This ANSI/CAN/UL 2900-2-3 Standard is under continuous maintenance, whereby each revision is approved in compliance with the requirements of ANSI and SCC for accreditation of a Standards Development Organization. In the event that no revisions are issued for a period of four years from the date of publication, action to revise, reaffirm, or withdraw the standard shall be initiated.

In Canada, there are two official languages, English and French. All safety warnings must be in French and English. Attention is drawn to the possibility that some Canadian authorities may require additional markings and/or installation instructions to be in both official languages.

Comments or proposals for revisions on any part of the Standard may be submitted to ULSE at any time. Proposals should be submitted via a Proposal Request in the On-Line Collaborative Standards Development System (CSDS) at <http://csds.ul.com>.

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This Edition of the Standard has been formally approved by the Technical Committee (TC) on Software Cybersecurity for Security and Life Safety Signaling Systems, TC 2900-2-3.

This list represents the TC 2900-2-3 membership when the final text in this standard was balloted. Since that time, changes in the membership may have occurred.

TC 2900-2-3 Membership

Name	Representing	Interest Category	Region
Ahmadi, Mike	M. Ahmadi	General	USA
Barker, David	Representing Self	General	USA
Baroncini, Vincent	Siemens Industry Inc	Producer	USA
Biggs, Douglas	UL Solutions	Testing & Stds Org	USA
Callahan, Haley	UL Standards & Engagement	TC Project Manager – Non-voting	USA
Chevalier, Mathieu	Genetec	Producer	Quebec, Canada
Cosman, Eric	OIT Concepts LLC	Non-voting	USA
Curtis, Bill	Consortium for IT Software Quality (CISQ)	Testing & Stds Org	USA
Dawson, Joe	EWA-Canada (an INTERTEK Co)	Testing & Stds Org	Newfoundland, Canada

TC 2900-2-3 Membership Continued on Next Page

TC 2900-2-3 Membership Continued

Name	Representing	Interest Category	Region
Dischert, Larry	Johnson Controls, Inc / Building Solutions North America	Commercial / Industrial User	USA
Dutta, Ashim	Eaton India Innovation Center	Producer	India
Fields, Jennifer	UL Standards & Engagement	TC Chair – Non-voting	USA
Garvy, Patrick	Honeywell	Producer	USA
Greko, Joshua	Dice Corp	Producer	USA
Griffith, Steve	NEMA	Non-voting	USA
Hoskins, Bryan	Oklahoma State University	General	USA
Lee, Simon	US Consumer Product Safety Commission	Non-voting	USA
Li, Xiaodong	Lenovo (Beijing) LTD	Producer	China
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Mendoza, Ernesto	Signify North America Corporation	Commercial / Industrial User	USA
Milke, James	University of Maryland	General	USA
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Thayer, Rodney	Smithee Solutions LLC	General	USA
Toika, Michael	Addison Fire Protection District #1	AHJ	USA
Tran, Phat	BC Safety Authority	Non-voting	British Columbia, Canada
Vasserman, Eugene	Kansas State University	General	USA
Wang, Hui	CNCERT/CC	General	China
Wyman, Richard	CS 7 Consulting	General	USA

International Classification for Standards (ICS): 35.030, 35.110, 35.240.50, 35.240.80

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This Standard is intended to be used for conformity assessment.

The intended primary application of this standard is stated in its scope. It is important to note that it remains the responsibility of the user of the standard to judge its suitability for this particular application.

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INTRODUCTION

Note: This Standard for Software Cybersecurity for Network-Connectable Products, Part 2-3: Particular Requirements for Security and Life Safety Signaling Systems refers to the Standard for Software Cybersecurity for Network-Connectable Products, Part 1: General Requirements, UL 2900-1.

1 Scope

1.1 This security evaluation standard applies to the evaluation of security and life safety signaling system components. It applies to, but is not limited to, the following products:

- a) Alarm Control Units;
- b) Network-Based Intrusion Detection System;
- c) General Purpose Signaling Units;
- d) Digital Video Equipment and Systems;
- e) Mass Notification and Emergency Communication / Evacuation Equipment and Systems;
- f) Control servers;
- g) Alarm Automation System Software;
- h) Alarm Receiving Equipment;
- i) Anti-Theft Equipment;
- j) Automated Teller Machines;
- k) Fire Alarm Control Systems;
- l) Network Connected Locking Devices;
- m) Physical Security Information Management (PSIM) Systems;
- n) Smoke Control Systems;
- o) Smoke / Gas / CO Detection Devices;
- p) Audible and Visual Signaling Devices (fire and general signaling);
- q) Access Control Equipment and Systems; and
- r) Smart Locks.

1.2 This standard does not contain general requirements that are intended to address functional testing of the product unless expressly specified.

1.3 This standard also describes requirements for the product risk management process carried out by the vendor of the product, including a list of security controls that the product (or the vendor, as applicable) shall comply with unless a risk assessment done by the vendor shows that the risk of not implementing one of these security controls is acceptable.

2 Normative References

2.1 All references are for the current published version of the document unless stated otherwise.

Normative References are included in Section 2 of the Standard for Cybersecurity for Network-Connectable Products, Part 1: General Requirements, UL 2900-1.

3 Glossary

Glossary Terms are included in Section 3 of the Standard for Cybersecurity for Network-Connectable Products, Part 1: General Requirements, UL 2900-1.

4 General

4.1 This standard includes three levels of security requirements that are applicable to the product with an increasing level of security for higher levels. The levels and their description are defined in [Table 4.1](#).

Table 4.1

Level	Description
L1	Includes foundational cybersecurity testing requirements for security risk assessment of software in products covered in this standard. Provides assessment of general security capabilities of a product with limited knowledge of the internal security controls of the product. L1 does not require the submission of source code. This level is closest to “black box” testing. L1 is recommended as a minimum level of assessment.
L2	Includes L1 assessment and testing requirements and additional supplemental requirements for security risks assessment of software in products. Source code is tested at this level. Provides assessment of security capabilities of a product with knowledge of internal security controls of the product. Because specific protections for sensitive data are included at L2, this is the lowest level recommended for products sending, receiving, or processing sensitive data.
L3	Includes L1 and L2 assessment and testing requirements and additional supplemental requirements of the vendor process and management. Provides assessment of security capabilities of a product with knowledge of internal security controls of the product and knowledge of the business practices of the vendor to support the lifecycle of the product.

4.2 The product shall comply with the clauses identified in the tables of each section of this standard per the Level intended. The level intended will be marked with an X per the applicable clause. Where an X is not applied, the clause is not mandatory for the Level. In keeping with this approach, when a clause calls for compliance with another section of this standard, only sub-clauses for the level intended (marked with an “X”) are applicable.

DOCUMENTATION OF PRODUCT, PRODUCT DESIGN AND PRODUCT USE

5 Product Documentation

Table 5.1

Clause	L1	L2	L3
5.1 <i>Deleted</i>			
5.2 The product shall comply with: Product Documentation, Section 4.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
5.3 The product shall comply with: Product Documentation, Section 4.1.1(b) of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

6 Product Design Documentation

Table 6.1

Clause	L1	L2	L3
6.1 The product shall comply with: Product Design Documentation, Section 5.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

7 Documentation for Product Use

Table 7.1

Clause	L1	L2	L3
7.1 The product shall comply with: Documentation for Product Use, Section 6.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
7.2 The product shall comply with: Documentation for Product Use, Section 6.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
7.3 The product shall comply with: Documentation for Product Use, Section 6.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
7.4 The product shall comply with: Documentation for Product Use, Section 6.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
7.5 The product shall comply with: Documentation for Product Use, Section 6.5 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	

Table 7.1 Continued on Next Page

Table 7.1 Continued

Clause	L1	L2	L3
7.6 The product shall comply with: Documentation for Product Use, Section 6.6 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
7.7 The product shall comply with: Documentation for Product Use, Section 6.7 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
7.8 The product shall comply with: Documentation for Product Use, Section 6.8 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
7.9 The product shall comply with: Documentation for Product Use, Section 6.9 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
7.10 The product shall comply with: Documentation for Product Use, Section 6.10 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	

RISK CONTROLS

8 General

Table 8.1

Clause	L1	L2	L3
8.1 The product (or the product's vendor, as applicable) shall comply with all of the applicable controls specified in Clauses 9 – 12 of this standard, unless the risk assessment performed by the vendor according to Section 13, Vendor Product Risk Management Process, shows that the risks associated with not implementing a specific control are acceptable in product use.	X		
8.2 The product shall comply with: Risk Controls – General, Section 7.1.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
8.3 The product shall comply with: Risk Controls – General, Section 7.1.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
8.4 The product shall comply with: Risk Controls – General, Section 7.1.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
8.5 The product shall comply with: Risk Controls – General, Section 7.1.5 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

9 Access Control, User Authentication and User Authorization

Table 9.1

Clause	L1	L2	L3
9.1 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.2 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.3 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.4 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.5 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.5 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.6 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.6 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.7 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.7 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.8 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.8 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
9.9 The product shall comply with: Access Control, User Authentication and User Authorization, Section 8.9 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

10 Remote Communication

Table 10.1

Clause	L1	L2	L3
10.1 The product shall comply with: Remote Communication, Section 9.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

11 Sensitive Data

Table 11.1

Clause	L1	L2	L3
11.1 The product shall comply with: Sensitive Data, Section 10.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
11.2 The product shall comply with: Sensitive Data, Section 10.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
11.3 The product shall comply with: Sensitive Data, Section 10.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
11.4 The product shall comply with: Sensitive Data, Section 10.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	

12 Product Management

Table 12.1

Clause	L1	L2	L3
12.1 The product shall comply with: Product Management, Section 11.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.2 The product shall comply with: Product Management, Section 11.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.3 The product shall comply with: Product Management, Section 11.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.4 The product shall comply with: Product Management, Section 11.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.5 The product shall comply with: Product Management, Section 11.5 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.6 The product shall comply with: Product Management, Section 11.6 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
12.7 The product shall comply with: Product Management, Section 11.7 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	
12.8 The product shall comply with: Product Management, Section 11.8 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.		X	

RISK MANAGEMENT

13 Vendor Product Risk Management Process

Table 13.1

Clause	L1	L2	L3
13.1 The product shall comply with: Vender Product Risk Management Process, Section 12.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
13.2 The product shall comply with: Vender Product Risk Management Process, Section 12.2 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
13.3 The product shall comply with: Vender Product Risk Management Process, Section 12.3 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
13.4 <i>Deleted</i>			
13.5 The product shall comply with: Vender Product Risk Management Process, Section 12.5 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
13.6 <i>Deleted</i>			
13.7 The product shall comply with: Vender Product Risk Management Process, Section 12.7 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
13.8 The product shall comply with: Vender Product Risk Management Process, Section 12.8 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	

SOFTWARE COMPOSITION

14 Known Vulnerability Testing

Table 14.1

Table deleted

14A Software Composition Analysis

Table 14A.1

Clause	L1	L2	L3
14A.1 The product shall comply with: Software Composition Analysis, Section 13A.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	

15 Malware Testing

Table 15.1

Clause		L1	L2	L3
15.1	The product shall comply with: Malware Testing, Section 14.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
15.2	The product shall comply with: Malware Testing, Section 14.2 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	

16 Malformed Input Protocol Testing (also reference Appendix D)

Table 16.1

Clause		L1	L2	L3
16.1	The product shall comply with: Malformed Input Testing, Section 15.1 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.2	The product shall comply with: Malformed Input Testing, Section 15.2 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.3	The product shall comply with: Malformed Input Testing, Section 15.3 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.4	The product shall comply with: Malformed Input Testing, Section 15.4 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.5	The product shall comply with: Malformed Input Testing, Section 15.5 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.6	The product shall comply with: Malformed Input Testing, Section 15.6 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.7	The product shall comply with: Malformed Input Testing, Section 15.7 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.8	The product shall comply with: Malformed Input Testing, Section 15.8 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.9	The product shall comply with: Malformed Input Testing, Section 15.9 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

Table 16.1 Continued on Next Page

Table 16.1 Continued

Clause	L1	L2	L3
16.10 The product shall comply with: Malformed Input Testing, Section 15.10 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		
16.11 The product shall comply with: Malformed Input Testing, Section 15.11 of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.	X		

17 Structured Penetration Testing**Table 17.1**

Clause	L1	L2	L3
17.1 The product shall comply with: Structured Penetration Testing, Section 16.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
17.2 The product shall comply with: Structured Penetration Testing, Section 16.2 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
17.3 The product shall comply with: Structured Penetration Testing, Section 16.3 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		
17.4 The product shall comply with: Structured Penetration Testing, Section 16.4 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.	X		

SOFTWARE WEAKNESS ANALYSIS**18 Software Weakness Analysis****Table 18.1**

Clause	L1	L2	L3
18.1 The product shall comply with: Software Weakness Analysis, Section 17.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
18.2 The product shall comply with: Software Weakness Analysis, Section 17.2 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
18.3 The product shall comply with: Software Weakness Analysis, Section 17.3 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	

19 Static Code Analysis

Table 19.1

Table deleted

SOFTWARE VULNERABILITIES

20 Static Binary and Bytecode Analysis

Table 20.1

Clause	L1	L2	L3
20.1 The product shall comply with: Static Binary and Bytecode Analysis, Section 19.1 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
20.2 The product shall comply with: Static Binary and Bytecode Analysis, Section 19.2 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
20.3 The product shall comply with: Static Binary and Bytecode Analysis, Section 19.3 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
20.4 The product shall comply with: Static Binary and Bytecode Analysis, Section 19.4 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	
20.5 The product shall comply with: Static Binary and Bytecode Analysis, Section 19.5 of the Standard for Software Cybersecurity for Network-Connectable Devices: General Requirements, UL 2900-1.		X	

21 Organizational Assessment

Table 21.1

Clause	L1	L2	L3
21.1 The vendor has a demonstrated process to track and identify vulnerabilities found in the software components of the product. This process has a mechanism to score or weight newly found vulnerabilities and communicate to the supply chain the existing problem and announcements of fixes.			X
21.2 The product is updated on a regular schedule. These updates are either automatic or there will be notifications set up for the end-user to update the product manually. The manual process of updating the product shall be straightforward and executed with ease.			X

Table 21.1 Continued on Next Page

Table 21.1 Continued

Clause	L1	L2	L3
21.3 There is a system in place to notify the appropriate authorities and affected users in the event of a data breach or any other severe cyber-attacks on the product and its related services.			X
21.4 The vendor will maintain the intended functionality of the product for its expected lifetime. This is done via customer support, updates, and not disabling certain features during the timeframe of the product's lifetime unless those features pose a risk to the product. If a feature or service must be removed, a replacement feature or service will be provided by the vendor.			X

APPENDIX A

A1 Sources for Software Weaknesses

Refer to Appendix A, Sources for Software Weaknesses, of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.

A2 Weaknesses and Vulnerabilities

Refer to Appendix A2, Weaknesses and Vulnerabilities, of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.

APPENDIX B

B1 Requirements for Secure Mechanisms for Storing Sensitive Data and Personally Identifiable Information

Refer to Appendix B, Requirements for Secure Mechanisms for Storing Sensitive Data and Personally Identifiable Information, of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.

APPENDIX C

C1 Requirements for Security Functions

Refer to Appendix C, Requirements for Security Functions, of the Standard for Software Cybersecurity for Network-Connectable Devices, Part 1: General Requirements, UL 2900-1.

APPENDIX D

D1 Level 1 Malformed Input Protocol List

- Network Protocols List
- IPV4, IPV6 including multicast
- TCP, UDP, ICMP
- HTTP
- HTTPS (HTTP over TLS)
- SMTP
- SMTP with STARTTLS (SMTP over TLS)
- DNS
- mDNS/ZeroConf
- UPNP
- EAPOL (protocol inside 802.1X)
- RADIUS
- SNMP
- SYLOG
- NTP including secure NTP
- Video Protocols List
- RTSP
- H.264, H.265
- FTP



333 Pfingsten Road
Northbrook, Illinois 60062–2096
847.272.8800

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